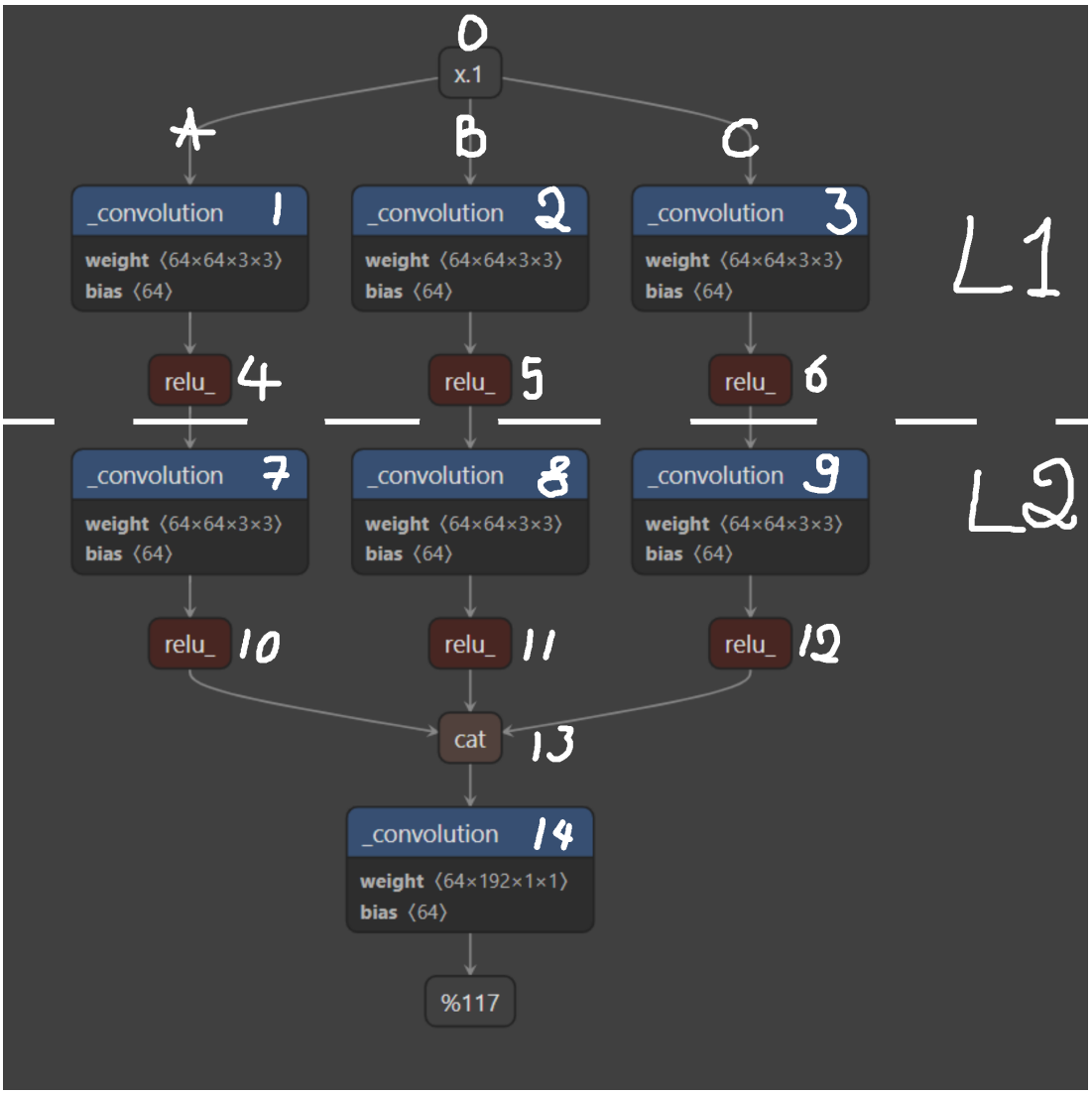
1 Cấu trúc DAG



Hình 1: DAG

Input: x.1 — shape (1, 64, 56, 56)

Tất cả Conv đều là stride=1, dilation=1, kernel=3

2 Stage 1: DAG Construction

2.1 Định danh các nodes

Node ID	Op type	Input tensors	Output tensor	Params
0	x.1 (input placeholder)	—	(1,64,56,56)	—
1	linalg.conv_2d_nchw_fchw (Branch A, L1)	x.1, W_A1	(1,64,56,56)	stride=1, dil=1
2	linalg.conv_2d_nchw_fchw (Branch B, L1)	x.1, W_B1	(1,64,56,56)	stride=1, dil=1
3	linalg.conv_2d_nchw_fchw (Branch C, L1)	x.1, W_C1	(1,64,56,56)	stride=1, dil=1
4	linalg.elemwise_unary / ReLU (Branch A)	T1	(1,64,56,56)	op=relu
5	linalg.elemwise_unary / ReLU (Branch B)	T2	(1,64,56,56)	op=relu
6	linalg.elemwise_unary / ReLU (Branch C)	T3	(1,64,56,56)	op=relu
7	linalg.conv_2d_nchw_fchw (Branch A, L2)	T4, W_A2	(1,64,56,56)	stride=1, dil=1
8	linalg.conv_2d_nchw_fchw (Branch B, L2)	T5, W_B2	(1,64,56,56)	stride=1, dil=1
9	linalg.conv_2d_nchw_fchw (Branch C, L2)	T6, W_C2	(1,64,56,56)	stride=1, dil=1
10	linalg.elemwise_unary / ReLU (Branch A)	T7	(1,64,56,56)	op=relu
11	linalg.elemwise_unary / ReLU (Branch B)	T8	(1,64,56,56)	op=relu
12	linalg.elemwise_unary / ReLU (Branch C)	T9	(1,64,56,56)	op=relu
13	tensor.concat / cat	T10, T11, T12	(1,192,56,56)	dim=1
14	linalg.conv_2d_nchw_fchw (1×1 conv)	T13, W_fin	(1,64,56,56)	stride=1, dil=1

2.2 Topological Order (dùng Kahn's + min-heap)

Topological order: [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14]

3 Stage 2: Liveness Analysis

3.1 Tính kích thước từng tensor

Ta có Float32 = 4 bytes/element

Size of Tensor:

T_input (x.1): 1x64x56x56 x 4 = 802,816 bytes ~ 784 KB
T1 = T2 = T3: 1x64x56x56 x 4 = 802,816 bytes ~ 784 KB
T4 = T5 = T6: 1x64x56x56 x 4 = 802,816 bytes
T7 = T8 = T9: 1x64x56x56 x 4 = 802,816 bytes
T10 = T11 = T12: 1x64x56x56 x 4 = 802,816 bytes
T13 (cat output): 1x192x56x56 x 4 = 2,408,448 bytes ~ 2.3 MB
T14 (final output): 1x64x56x56 x 4 = 802,816 bytes

Weights:

W_Ax / W_Bx / W_Cx (L1, L2): 64x64x3x3 x 4 = 147,456 bytes
W_fin: 64x192x1x1 x 4 = 49,152 bytes

3.2 Tính $p(t)$ và $\ell(t)$ cho mỗi tensor trung gian

Tensor	Producer (p)	Consumers	$\ell(t) = \max(\textbf{consumers})$
T_input		{1, 2, 3}	3
T1	1	{4}	4
T2	2	{5}	5
T3	3	{6}	6
T4	4	{7}	7
T5	5	{8}	8
T6	6	{9}	9
T7	7	{10}	10
T8	8	{11}	11
T9	9	{12}	12
T10	10	{13}	13
T11	11	{13}	13
T12	12	{13}	13
T13	13	{14}	14
T14	14	{} (sink)	14

3.3 Bảng Live Tensors tại mỗi program point

Tensor t alive tại point i khi: $p(t) \leq i \leq \ell(t)$

Point i	Live Tensors	Tổng bytes
0	T_input	802,816
1	T_input, T1	1,605,632
2	T_input, T1, T2	2,408,448
3	T_input, T1, T2, T3	**3,211,264**
4	T_input, T1→dead, T2, T3, T4	3,211,264
5	T_input, T2→dead, T3, T4, T5	3,211,264
6	T_input, T3→dead, T4, T5, T6	3,211,264
7	T_input→dead, T4→dead, T5, T6, T7	2,408,448
8	T5→dead, T6, T7, T8	2,408,448
9	T6→dead, T7, T8, T9	2,408,448
10	T7→dead, T8, T9, T10	2,408,448
11	T8→dead, T9, T10, T11	2,408,448
12	T9→dead, T10, T11, T12	2,408,448
13	T10→dead, T11→dead, T12→dead, T13	2,408,448
14	T13→dead, T14	802,816

4 Stage 3: Candidate Selection

Budget = 512 KB × 80% = 409,600 bytes

Conv L1 (nodes 1, 2, 3)

op_type = conv_2d_nchw_fchw
output shape: (1, 64, 56, 56)
input shape: (1, 64, 56, 56)
weight shape: (64, 64, 3, 3)

Full working set:
output: 1x64x56x56x4 = 802,816 bytes
input: 1x64x56x56x4 = 802,816 bytes
weight: 64x64x3x3x4 = 147,456 bytes

Total: 1,753,088 bytes ~ 1.67 MB

1,753,088 > Budget -> Cắt

ReLU (nodes 4–6, 10–12)

Full working set = output + input = 2 x 802,816 = 1,605,632 bytes
1,605,632 > Budget -> Cắt

Conv L2 (nodes 7–9)

Total = 1,753,088 bytes > Budget -> Cắt

Cat (node 13)

tensor.concat, không phải linalg.
-> Hong cắt

Conv cuối (node 14)

Total: 3,260,416 bytes ~ 3.11 MB
3,260,416 > Budget -> Cắt

Danh sách candidates cuối

Node	Op	Full WS (bytes)	Cắt?
1, 2, 3	Conv 3×3 L1	1,753,088	✓
4, 5, 6	ReLU L1	1,605,632	✓
7, 8, 9	Conv 3×3 L2	1,753,088	✓
10, 11, 12	ReLU L2	1,605,632	✓
13	cat	N/A	×
14	Conv 1×1	3,260,416	✓

5 Stage 4: Dimension Selection

Ưu tiên cắt theo H trước cho cả Conv và ReLU

6 Stage 5: Tile Size Calculation

Budget = 409,600 bytes

Conv L1, L2

shape = (1, 64, 56, 56), dim = spatial_h, stride=1, dilation=1, kernel=3

Output tile:

(1, 64, T_h, 56) = 64 * T_h * 56 * 4 = 14,336 * T_h bytes

Ta có:

h_in = (T_h - 1) * stride + dilation * (kernel_h - 1) + 1
= (T_h - 1) * 1 + 1 * 2 + 1
= T_h + 2 (Thêm 2 rows halo)

Input tile:

(1, 64, T_h+2, 56) = 64 * (T_h+2) * 56 * 4 = 14,336 * (T_h+2) bytes

Weight tile:

(64, 64, 3, 3) * 4 = 147,456 bytes

Total footprint:

F(T_h) = 14,336*T_h + 14,336*(T_h+2) + 147,456
= 28,672*T_h + 28,672 + 147,456
= 28,672*T_h + 176,128

Từ đầy ta tính được T_h = 8 là tối ưu nhất:

F(8) = 28,672*8 + 176,128 = 229,376 + 176,128 = 405,504 <= Budget

Với T_h = 8**, ta có**
Mỗi tile = 8 rows output, cần 10 rows input (8 + 2 halo).
Số tiles = ⌈56/8⌉ = **7 tiles**

Tất cả ReLU

Tương tự:

F(T_h) = 28,672 x T_h
best: T_h = 14 -> align to 8

Tile size cho Conv cuối: T_h = 4
Số tiles = ⌈56/8⌉ = **7 tiles**

Conv cuối

Tương tự:

F(T_h) = 57,344 x T_h + 49,152
best: T_h = 6 -> align to 4

Tile size cho Conv cuối: T_h = 4
Số tiles = ⌈56/4⌉ = **14 tiles**

7 Stage 6: Schedule Search

Vì chỉ có **1D cascade** và 1 dimension hợp lệ cho mỗi op:

Node	Op	Slice dim	Tile size	Tiles
1, 2, 3	Conv3×3 L1	H	**8**	7
4, 5, 6	ReLU L1	H	**8**	7
7, 8, 9	Conv3×3 L2	H	**8**	7
10, 11, 12	ReLU L2	H	**8**	7
14	Conv1×1	H	**4**	14

8 Stage 7: Fusion Planning

8.1 Fusion chains

Final anchors:

{1, 2, 3} (Conv L1 mỗi nhánh)
{7, 8, 9} (Conv L2 mỗi nhánh, ReLU L1 sẽ fuse vào đây)
{10, 11, 12} (ReLU L2)
{14} (Conv 1x1)

8.2 Re-fit tile sau fusion

Group members:

Node 7 (ConvL2) + Node 4 (ReLU_A1)

Group total:

F_group(T_h) = 57,344 x T_h + 176,128
best: T_h = 4 -> align to 4

Tile size sau fusion:

Node	Op	Slice dim	Tile size	Tiles
1, 2, 3	Conv3×3 L1	H	**8**	7
7, 8, 9	Conv3×3 L2 fused	H	**4**	14
10, 11, 12	ReLU L2	H	**8**	7
14	Conv1×1	H	**4**	14

9 Stage 8: MLIR Transform IR Generation

Cái này hong có tính được

10 Memory tiers

Op **Memory tier**	**Working set tile**
Conv L1 (tile H=8)	405,504 B ≈ 396 KB
Conv L2 fused (tile H=4)	405,504 B ≈ 396 KB
ReLU L2 (tile H=8)	$14,336 \times 8 \times 2 = 229,376$ B
Conv 1×1 (tile H=4)	393,216 B ≈ 384 KB